Lesson notes · for you

A picture is a grid of numbers

University, vision 1

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- One number per cell
- Three grids for colour

Activity: In the lab

1. Pick Upright edges and press “Slide the filter”. Read one sum under “At this spot”.
2. Switch to Blur, then Sharpen, and compare the new pictures with the first one.
3. Write one sentence on what none of the filters knows about the picture.

Discussion

- What is the filter mixing?

Check yourself

- A filter does not know the name of what?

Try it this week
Sketch a 3×3 upright-edge filter and one output pixel by hand.

Worksheet

A picture is a grid of numbers

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Name | Class | Date

1 **The upright-edge filter sits on this patch. Multiply each cell by its weight and add. What is the new pixel?**

```
patch  = [[0, 0, 9],
          [0, 0, 9],
          [0, 0, 9]]
kernel = [[-1, 0, 1],
          [-2, 0, 2],
          [-1, 0, 1]]
```

◯ 27 ◯ 0 ◯ 36

2 **A 3×3 filter mixes each pixel with its neighbours; it does not know it is looking at a face.**

◯ True ◯ False

3 **Match the filter.**

Upright edges	• • • •	A pixel averaged with its neighbours
Blur	• • • •	Keep the strongest in a small block
Sharpen	• • • •	The sides of walls and doors
Pooling	• • • •	A pixel made to stand out from its neighbours

4 **Order one filter step.**

☐ Multiply each cell by its weight, then add

☐ Write that sum as the new pixel

☐ Lay the 3×3 on the picture

✎ **What did the filter not know?**

Check your answers

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1 **36**

The left column of the patch is 0 and the middle weights are 0, so only the right column counts: 9 + 18 + 9 = 36.

2 **True**

It only multiplies nine cells by its weights and adds them.

3 **Upright edges — The sides of walls and doors**
Blur — A pixel averaged with its neighbours
Sharpen — A pixel made to stand out from its neighbours
Pooling — Keep the strongest in a small block

Convolution mixes. Pooling throws detail away.

4 **Top to bottom: 2, 3, 1**

The new picture is those sums, one per position.